Microfluidic Cell-viability Monitoring Device based on Sharp-edge Acoustic Streaming

[1]Admas Dagnachew Temesgen, [1]Minjoon Jeon, [1]Sofonias Nursefa Kedir, [2] Hyungjoon Ryu, [2]Hoyoung Choi, [2]Myunghee Kim, [2]Joonhui Kim, [1]Dohyun Kim *
[1]Myongji University, [2]LTIS Co., Ltd.
*E-mail: dohyun.kim@mju.ac.kr

샤프에지 음향 유동 기반 미소유체 세포 생존율 모니터링 디바이스

[1] 아드마스 다그나츄 테메스젠, [1] 전민준 [1] 케드르 소포니아스 눌세파, [2] 유형준, [2] 최호영, [2] 김명희, [2] 김준휘, [1] 김도현*
[1] 명지대학교, [2]㈜엘티아이에스

Abstract

We *for the first time* report a novel cell-viability monitoring microdevice based on sharp-edge acoustic streaming. Conventional, macroscopic approaches based on Trypan Blue (TB) exclusion are reagent- and time-consuming, lacking the continuous viability monitoring capability which is essential to various biomedical applications. To address these limitations, we designed a microfluidic device incorporating sharp-edge structures and fabricated it using a DLP 3D printer. Strong acoustic streaming generated by oscillating sharp edges enabled effective on-chip Trypan Blue (TB) staining of flowing cells. First, we compared TB exclusion assays with different mixing conditions to establish a standard protocol and analyze the impact of shear stress and mixing time on cell viability. The impact of the temperature on viability was also analyzed. Operating parameters, including excitation frequency, voltage, and flow rate, were systematically optimized through high-speed flow visualization to enhance streaming performance while maintaining device integrity. Mixing efficiency was also evaluated using fluorescent dye across a range of flow rates. Under optimized operating conditions, we successfully measured K562 cell viability, yielding a value similar to that of the conventional TB assay.

Keywords: ***Trypan Blue exclusion assay, Micromixer, Sharp-edge acoustic streaming, Cell viability, Flow visualization***

1. Introduction

Cell viability assessment, which provides cellular-health information, is essential across various applications, including routine cell culture, cytotoxicity tests, and bioreactor monitoring. Conventional batch-type assays based on Trypan-Blue (TB) exclusion require frequent sampling and manual mixing of dye and cells, rendering them unsuitable for continuous monitoring [1].

TB exclusion requires mixing dye with cells. However, at the microscale, effective mixing is challenging due to the low Reynolds number (Re<1) [2]. Acoustofluidic micromixers have attracted significant attention due to rapid mixing and simple microfabrication. Recently, sharp-edge acoustic streaming has been widely employed due to stable streaming [3]. We *for the first time* propose a microfluidic device for continuous cell-viability monitoring based on sharp-edge acoustic streaming (Fig. 1a).

First, TB exclusion assays with different mixing conditions are compared to establish a standard protocol and analyze the impact of shear stress and mixing time. The impact of temperature was also analyzed to optimize the acoustic excitation condition. Secondly, we designed a device and fabricated it using a DLP 3D printer. Its operating condition was then optimized for effective and stable mixing, and the mixing performance was characterized. Lastly, the cell viability measured via our device was validated with a control.

2. Experimental

2.1 Design and fabrication

The microdevice has two inlets and one outlet, and six sharp-edge structures on both sidewalls of a central microchannel. The device was fabricated using a high-resolution DLP 3D printer (Asiga MAX X27) with PR-48 resin. After post-printing process, including UV curing and washing, surface treatment was undertaken to improve the optical transparency. Finally, a PZT actuator (~6.3 kHz resonance) and Luer adapters were bonded to the device using superglue to complete the fabrication (Fig. 1b).

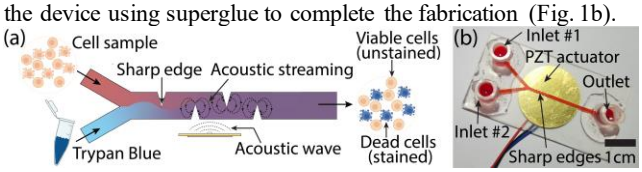

Fig. 1. (a) Conceptual diagram and (b) photograph of our microfluidic cell-viability monitoring device.

2.2 Study of Trypan Blue exclusion assays

As a control, cell viability was evaluated using the standard TB exclusion assay using K562 cells. The effects of shear stress and staining time on viability were examined under three mixing conditions: standard protocol (brief pipetting and incubation), repetitive pipetting, and vortexing. Stained (dead) and unstained (live) cells were counted using a hemocytometer (NanoEnTek) under a microscope (Olympus). To assess the influence of temperature, cell suspensions were preheated to temperatures in a range of 25 - 50°C for 1 h using a heating block prior to staining.

2.3 Mixing conditions optimization

The resonance frequencies of the device were identified using an impedance analyzer (Zurich Instruments) to excite the PZT actuator. Acoustic streaming performance was analyzed using high-speed flow visualization with fluorescent tracer microbeads (6.11 µm, Spherotech) and a high-speed sCMOS camera (PCO). The voltage

generating the fastest and broadest streaming was determined in a range of 20-80 V_{pp}. The excitation frequency producing a strong and stable streaming was determined among the identified resonance frequencies (3.38, 7.56, and 11.9 kHz). The impact of flow rate on streaming was also analyzed in a range of 3-15 µL/min.

2.4 Mixing performance analysis

Mixing performance was analyzed by co-flowing Alexa 488 dye solution and DI water at total flow rates of 2, 6, 10, and 15 µL/min. All experiments were conducted at the previously optimized excitation condition. The mixing index (MI) was calculated from pixel-wise fluorescence intensities at six different positions along the channel to quantitatively evaluate mixing performance.

2.5 On-chip cell viability assay

The on-chip cell viability assay was performed and evaluated. The device was operated under optimized excitation conditions and a total flow rate of 12 µL/min. The stained cells were collected, and viability was measured using an off-chip hemocytometer (NanoEnTek) and compared against an off-chip control.

3. Result and discussion

3.1 Study of Trypan Blue exclusion assays

Under the standard protocol (pipetting 2 times and incubating for 3 min), viability was 89.4% (control) at ~180 s. The viability was overestimated at shorter staining times (<40 s) due to limited mass transfer and underestimated at longer staining times (>180 s) due to the TB cytotoxicity (Fig. 2a). Repetitive (5 times) pipetting (Fig. 2b, optimal time = ~5 s) and vortexing (Fig. 2c, optimal time = ~4 s) showed similar trends that insufficient mixing overestimated viability whereas prolonged mixing reduced viability due to cytotoxicity and shear stress (especially vortexing). Similarly, increasing temperature progressively decreased viability with a threshold temperature at 40 °C. From these results, we concluded that (1) enhanced mixing reduces staining time, but (2) excessive staining time, shear stress, and temperature significantly reduce cellular health leading to cell-viability underestimation.

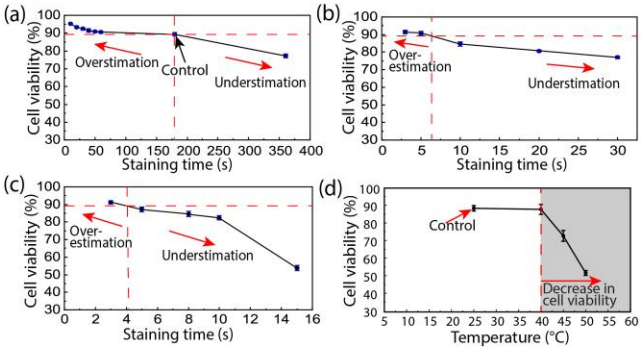

Fig. 2. (a–c) Cell viability under different mixing conditions. (d) Effect of temperature on cell viability.

3.2 Mixing conditions optimization

The fastest and widest streaming was observed at 80 V_{pp} (Fig. 3a). Higher voltages (≥100 V_{pp}) caused the temperature to exceed the threshold 40 °C. Excitation at the lowest bending mode (3.7 kHz) produced stable and strong acoustic streaming, whereas higher frequencies generated stronger but inconsistent streaming. At these optimized conditions (3.7 kHz, 80 V_{pp}), increasing the flow rate Q weakened streaming as eddies around the oscillating sharp edges are progressively suppressed by incoming flows. The strongest streaming was observed at the slowest 3 µL/min.

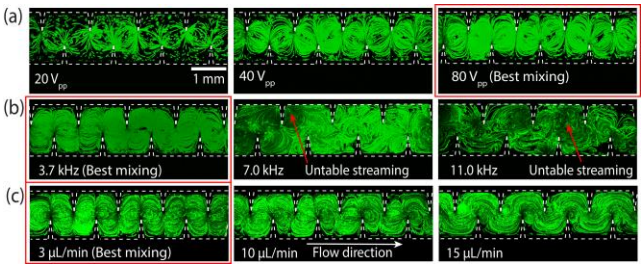

Fig. 3. Optimization of (a) excitation voltage, (b) excitation frequency, and (c) flow rate.

3.3 Mixing performance analysis

Both fluorescence images and MI graphs (Fig. 3a, b) indicate that mixing was completed at measurement positions #2, #4, and #6 for Q = 2, 6, and 10 µL/min, respectively. At 15 µL/min, mixing was incomplete, as indicated by non-uniform fluorescence intensity across the channel. As also shown in Fig. 3, mixing performance was degraded with increasing Q. The maximum usable flow rate was 10 µL/min.

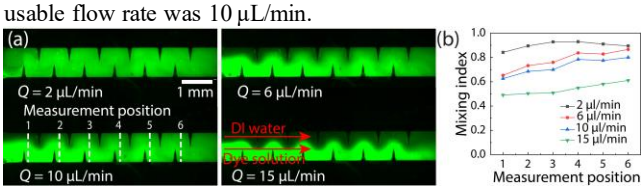

Fig. 3. (a) Mixing of fluorescence dye solution and DI water at different flow rates and (b) corresponding mixing index (MI) graph.

3.4 On-chip cell viability assay

On-chip assay shows a viability of 72.2 %, with a slight decrease from the control (78.4 %, Fig. 5), indicating that shear stresses and thermal effects from acoustic streaming were minimal. Hemocytometer images showed a clear distinction between viable (unstained) and non-viable (stained) cells.

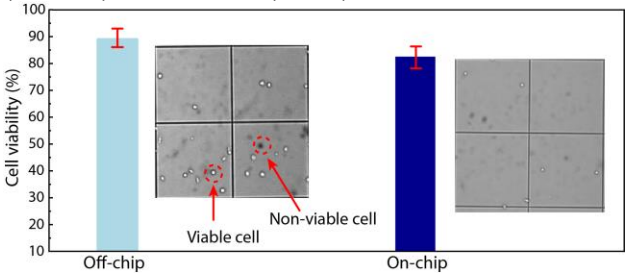

Fig. 5. Cell viability from off-chip control (78.4 %) vs. on-chip assay (72.2 %) and corresponding stained/unstained cell images.

4. Conclusion

We *for the first time* demonstrated a microfluidic cell-viability monitoring device based on sharp-edge acoustic streaming. The optimized operating conditions allowed viability measurement close to the control data obtained using a standard TB exclusion assay. We expect our device to be widely used for continuous cellular health monitoring in various biomedical applications.

Acknowledgments

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